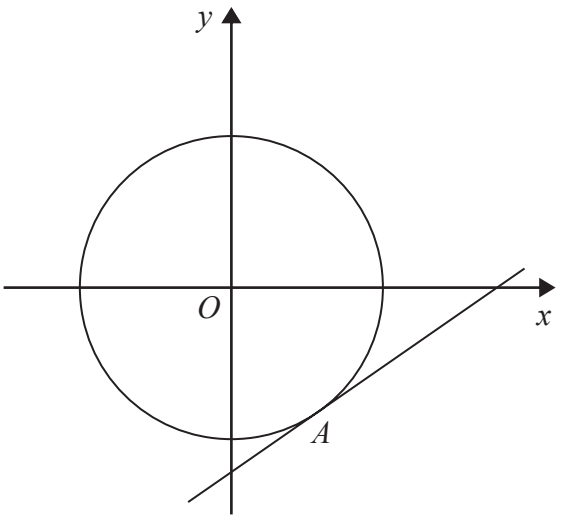
22 The diagram shows the circle with equation $x^2 + y^2 = 261$



A tangent to the circle is drawn at point A with coordinates $(p, -15)$, where $p > 0$

Find an equation of the tangent at A.

$p^2 + (-15)^2 = 261 \rightarrow p^2=36 \rightarrow p=6$ $(p>0)$.

1 mark
Find p=6

Gradient OA = $-15 \div 6 = -5/2$.

1 mark
Radius gradient

Tangent perpendicular to radius: $m = 2/5$.

1 mark
Negative reciprocal

Use $y = 0.4x + c$ through A(6,−15).

$-15 = 0.4\times6 + c \rightarrow c = -17.4$.

1 mark
Use point A

$y = 0.4x - 17.4$

1 mark
y=0.4x−17.4

(Total for Question 22 is 5 marks)